Topology-resolved quantum volume

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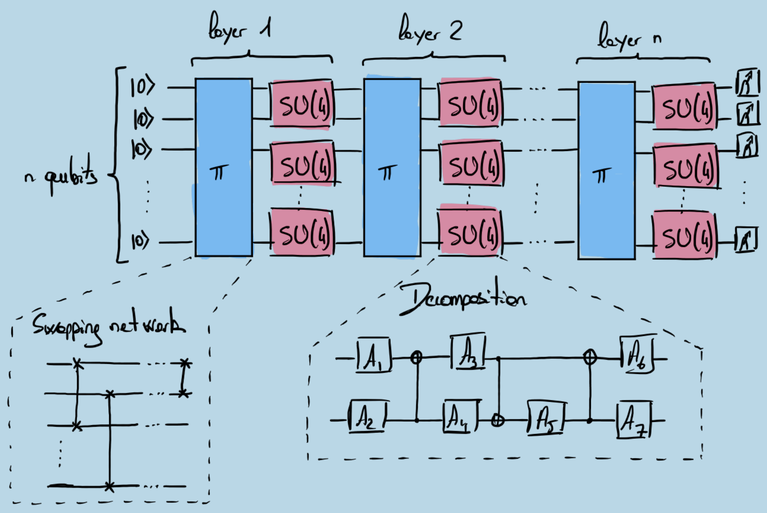
Problem

- NISQ devices have useful scale, but are limited by noise and lack fault tolerance.
- Performance depends on qubit quality, connectivity, and compilation overhead.
- Therefore, system-level benchmarks are needed in addition to gate-level metrics.

One such benchmark is quantum volume, which assesses only a limited subset of best-quality qubits within a quantum processor. We study how quantum volume depends on *where* a circuit is placed on a chip, not only on the chip as a whole.

Quantum volume

Quantum volume (QV) is a **system level benchmark**. The QV is a single number figure of merit used to evaluate universal gate-based quantum computers.



The QV circuit [1]

If the quantum computer samples a distribution from the **heavy set of outputs** (simulated classically) with probability $> 2/3$, it validates the corresponding quantum volume score of 2^n. The confidence interval for this evaluation is set to 2σ (97.7%).

Heavy set of outputs

Let $\mathbf{Q}$ be a random circuit drawn from a suitable ensemble acting on n qubits. The quantum state after executing the circuit is denoted $|\psi\rangle$. Each possible output state $x \in \{0,1\}^n$ is measured with probability $|\langle x|\psi\rangle|^2$. The set of output states with a probability greater than the median constitutes the heavy set of outputs associated with the quantum circuit $\mathbf{Q}$.

References

[1] Quantum Benchmark Zoo: *Quantum Volume*. Available at: `https://quantumbenchmarkzoo.org` (accessed 2026-04-12).

Acknowledgements

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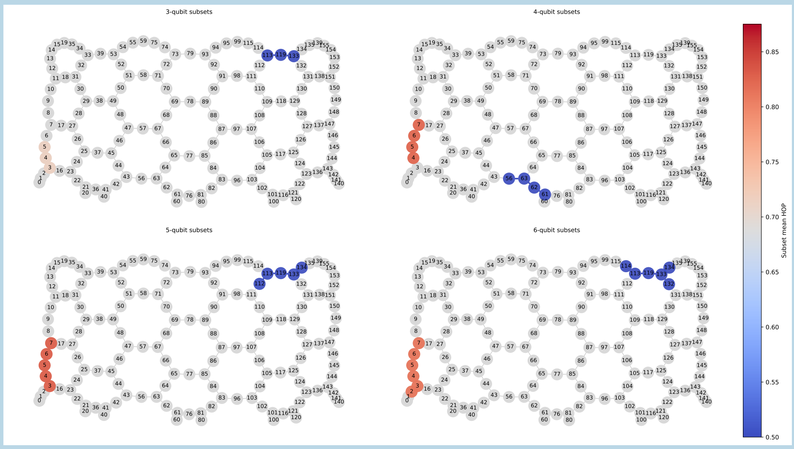
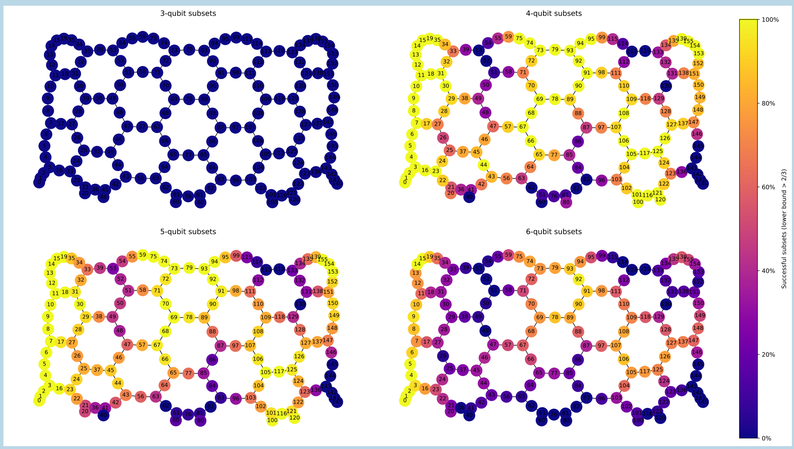
Research methodology

We benchmarked QV on the IBM Marrakesh device using calibration data and the coupling graph from Qiskit. First, we construct a noisy quantum simulator from the extracted data. Then we enumerate all connected subgraphs of size $n \leq 6$ and run QV circuits on each subgraph. Finally, we compute the heavy output probability and confidence for each placement.

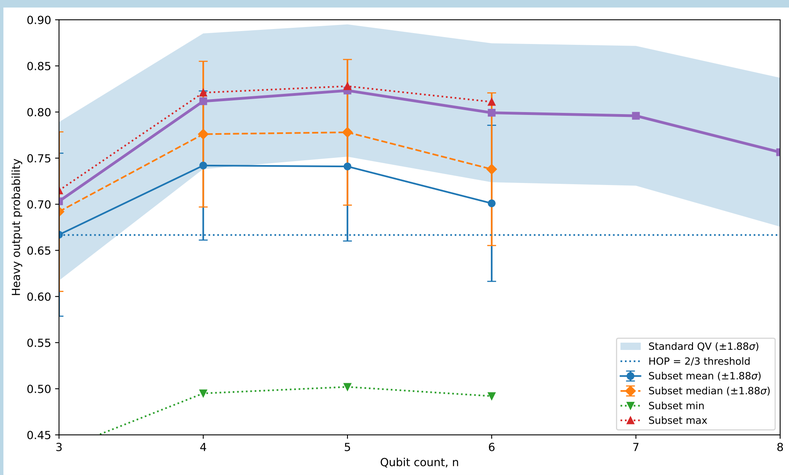
Productivity of qubit subsets varies significantly



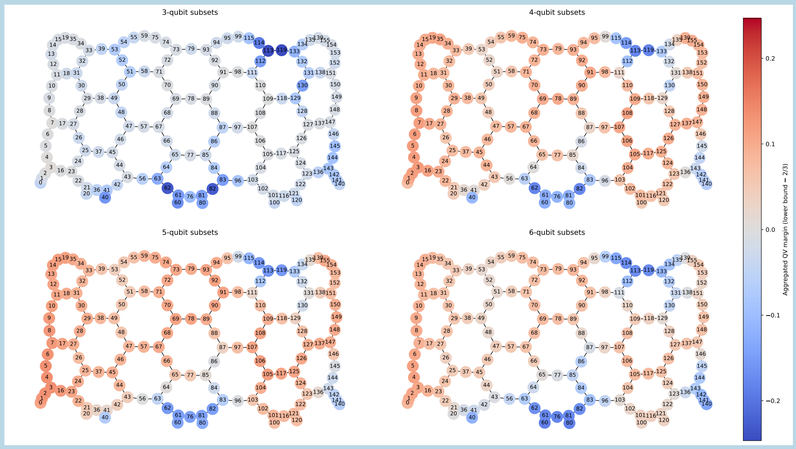
The best and worst HOP wise performing subsets of different subset sizes. Comparison makes the topology dependence clear: some regions of the chip are much better than others.



Empiric probability that a qubit is part of a subset that passes the QV test over different subset sizes. The success-fraction plot highlights that not all connected subsets of the same size perform equally well.



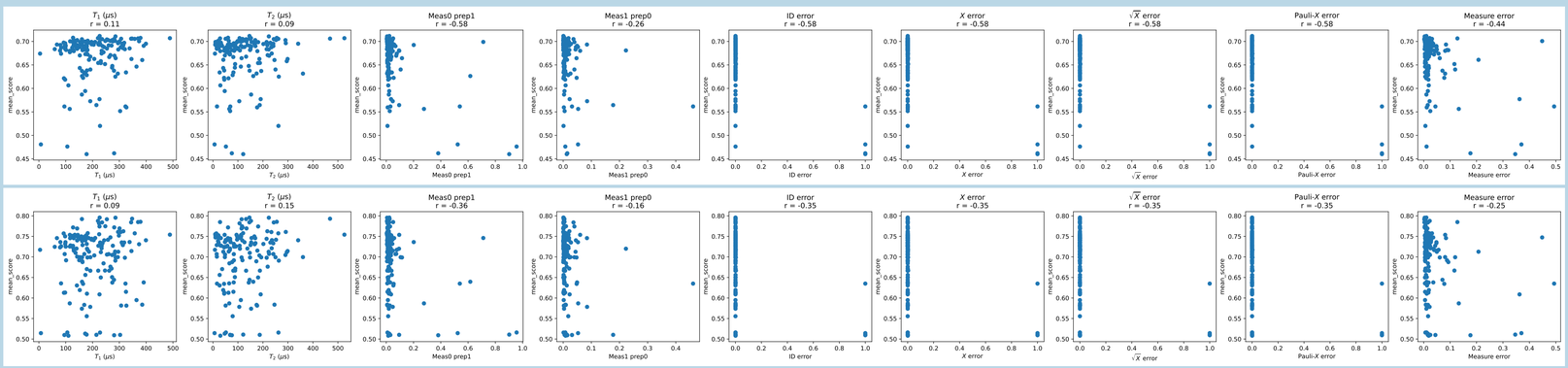
Statistics of all subsets of different sizes, showing the mean and 97% confidence interval of the heavy output probability.



Aggregated QV margin for each qubit, computed from all subsets containing it. Each node participates in many connected subsets. For a given node i, we aggregate heavy output probabilities $\{\mu_j\}$ from all subsets j containing i. **Mean score** is calculated as follows: $\bar{\mu}_i = \frac{1}{k_i}\sum_{j=1}^{k_i} \mu_j$. **Mean error** assumes independence $\sigma_i = \sqrt{\sum_{j=1}^{k_i} \sigma_j^2} \,/\, k_i$. **Lower confidence bound** takes into account the confidence interval of 2σ. Finally, we compute the **QV margin** by subtracting the QV threshold of $\frac{2}{3}$ from the lower confidence bound. Note that subsets overlap, so errors are not fully independent; the uncertainty is therefore slightly underestimated.

	3	4	5	6
T_1 (μs)	0.11	0.10	0.08	0.09
T_2 (μs)	0.09	0.13	0.13	0.15
Meas0 prep1	-0.58	-0.47	-0.41	-0.36
Meas1 prep0	-0.26	-0.20	-0.17	-0.16
ID error	-0.58	-0.46	-0.40	-0.35
X **error**	-0.58	-0.46	-0.40	-0.35
$\sqrt{X}$ **error**	-0.58	-0.46	-0.40	-0.35
Pauli-X error	-0.58	-0.46	-0.40	-0.35
Measure error	-0.44	-0.34	-0.29	-0.25

Correlation of qubit-level metrics with the aggregated QV margin for each qubit.



Correlation of qubit-level metrics with the aggregated QV margin for each qubit. The first line conatins correlations for subsets of size 3, the second line for subsets of size 6. The correlation values are generally higher for smaller subsets, which is expected as they are more sensitive to noise and errors. The most strongly correlated metrics are readout error, measurement preparation errors, and gate errors, while coherence times (T_1 and T_2) show weaker correlations.